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Professional Summary

Applied AI / Deep Learning practitioner with hands-on PyTorch experience training, evaluating, and benchmarking transformer-based and CNN-based models for multimodal and document-intelligence problems, alongside full-stack engineering skills (React, Node.js, FastAPI, Django) for shipping research into production systems. Experience spans cross-modal fusion, retrieval-augmented systems (RAG, LangGraph, ChromaDB), and rigorous evaluation methodology — comfortable training and debugging models, not just calling AI APIs, with a track record of owning problems end-to-end.

Education

B.Tech in Computer Science Engineering – SRM University AP CGPA: 8.13 *Aug 2023 – May 2027*

Technical Skills

Deep Learning & AI: PyTorch, Transformer Architectures, CNNs (ResNet-50/18), ArcFace, Siamese Networks, Model Training & Evaluation, Benchmarking, OpenCV, NLP, LangGraph, RAG (hybrid retrieval), Prompt/Agent Design

Languages: Python (Intermediate), JavaScript (ES6+), TypeScript, Java, C++ (basic)

Frontend/Mobile: React.js, HTML5, CSS3, Tailwind CSS, Flutter (basic)

Backend & APIs: Node.js, Express.js, FastAPI, Flask, Django, REST APIs, JWT Authentication, Microservices

Databases: MySQL, PostgreSQL, MongoDB, ChromaDB (vector), SQL Querying

Cloud & DevOps: AWS, Docker, Kubernetes (basic), Git, GitHub, CI-style workflow automation

Data & Tooling: Pandas, NumPy, Jest, PyTest

Soft Skills: Communication, Teamwork, Leadership, Problem-Solving

Experience

Virtual Research Intern – Universiti Teknologi PETRONAS, Malaysia [GitHub] *Apr 2025 – Jul 2025*

Dart, Flutter

- Identified and resolved asynchronous dictionary-loading issues causing UI freezes at startup, improving load reliability and creating a smoother first-run experience for users.
- Translated open-ended gameplay requirements into structured features: word validation, multiple difficulty levels, timed rounds, and scoring, documenting the implementation to support future development.
- Improved usability with responsive layouts and audio feedback, collaborating with the team to align the app with user needs.

Projects

Multimodal Biometric Authentication System [GitHub]

Python, PyTorch, ResNet-50, ArcFace, Siamese CNN, ResNet-18, Transformer Encoder, OpenCV, PyTest

- Designed, trained, and evaluated a multimodal deep learning system fusing face, fingerprint, and palmprint biometrics using ArcFace + ResNet-50 and Siamese CNN + ResNet-18 across 500 identities / 20,000 images.
- Built a transformer-based encoder to learn cross-modal fusion weights automatically, replacing manual tuning with a learned attention mechanism – direct hands-on experience with transformer architectures beyond tutorials.
- Ran systematic evaluation and benchmarking across 175,700 trials, achieving 99.99% accuracy, 0.0063% EER, and 0.9999 AUC, while optimizing the inference pipeline to under 1 second on GPU, outperforming every unimodal baseline.

Industrial Brain AI – Enterprise AI Knowledge Intelligence Platform [GitHub]

React.js, FastAPI, PostgreSQL, ChromaDB, LangGraph, Docker, Groq, Gemini

- Built a retrieval-aware, evaluation-driven AI system for document intelligence, combining hybrid RAG (SQL + vector + knowledge graph), LangGraph agents, and citation-based reasoning for industrial compliance and root-cause analysis workflows.
- Owned the system end-to-end – architecture, API design, and containerized deployment – while iterating on model/embedding pipeline changes (embedding-001 → gemini-embedding-001) and fixing agent response schema mismatches.
- Optimized the document-processing pipeline with compiled regex and LRU caching, and improved response latency via streaming and SQL-based search, applied to real-world industrial equipment manuals.

MockInterviewHub – AI Career Preparation Platform [GitHub]

React.js, Node.js, Express.js, MongoDB, Google Gemini AI, Groq SDK, JWT, Jest, PyTest

- Built a full-stack web application end-to-end with a React.js frontend and an Express.js/MongoDB backend, integrating Google Gemini AI and Groq SDK to power voice and written mock interviews, resume analysis, and skill-gap detection.
- Designed and built REST API endpoints with file-upload validation (multer), including PDF/DOCX type checks, a 10MB size limit, and magic-byte verification, eliminating malformed-file errors during testing.
- Implemented JWT authentication with bcrypt password hashing and modeled interview session history in MongoDB so users can track how their answers improve across rounds.

Certifications

- Oracle Certified Professional: Java SE 17 Developer** – Oracle | Java, OOP, and enterprise backend development.
- MongoDB Certified Associate Developer** – MongoDB | CRUD, indexing, aggregation pipelines, and scalable NoSQL schema design.

Achievements

- Led a four-person team end-to-end, from scoping to deployment, to build an AI-based waste management solution, finishing as a Top Finalist among 400+ teams in a 48-hour national hackathon.
- Ranked 541 in the PW Skills Coding Contest and solved 300+ Data Structures and Algorithms problems on LeetCode, demonstrating strong problem-solving and interview readiness.